

Topic 2

Using Mental Strategies to Solve Problems Involving Fractions Mathematics Discussion

In Unit 2, we learnt about partitioning fractions, equivalent fractions and counting forwards and backwards by unit fractions on a number line.

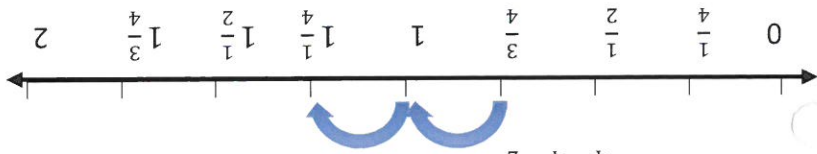
We can extend this thinking to be able to solve problems involving **addition** and **subtraction** of fractions. For example, we can solve a problem like $\frac{3}{4} + \frac{2}{1}$ by visualising:

1. A shape, such as a circle or rectangle, with $\frac{3}{4}$ shaded, with $\frac{2}{4}$ to it and then $\frac{1}{4}$ more (because adding $\frac{2}{2}$ is equivalent to adding $\frac{1}{1}$).



We can then see that $\frac{3}{4} + \frac{2}{1} = 1\frac{1}{4}$

2. A number line. We can start at $\frac{3}{4}$ and then count $\frac{2}{4}$ forward and then $\frac{1}{4}$ more (because $\frac{1}{1} + \frac{1}{4} = \frac{5}{4}$).



We can use the same visualisation of diagrams and number lines for problems involving subtraction of fractions.

In Unit 2 we learnt to partition the whole into parts in order to compare fractions. We can extend this thinking to help us solve problems involving **multiplication**.

For example, we can solve a problem like $\frac{3}{2} \times 9$ by drawing 9 objects.



We can partition these into 3 groups because finding $\frac{3}{1}$ of an amount is the same as dividing by 3.



$\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, 1\frac{1}{4}, 1\frac{2}{4}, 1\frac{3}{4}, 2, 2\frac{1}{4}, 2\frac{2}{4}, 2\frac{3}{4}, 3$

Continue shading to complete the fraction sequence for 3 wholes.

as $\frac{4}{4}$ or 1. Write how much of the circle is shaded in the sequence below.

Shade another quarter in the first circle in **green**. We have now shaded $\frac{4}{4} + \frac{1}{4}$ which is the same

as $\frac{5}{4}$. Write how much of the circle is shaded in the sequence below.

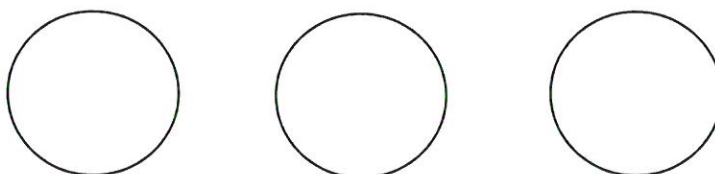
Shade another quarter in the first circle in **black**. We have now shaded $\frac{5}{4} + \frac{1}{4}$ which is the same

$\frac{6}{4}$. The amount shaded is shown in the sequence below.

Shade another quarter in the first circle in **blue**. We have now shaded $\frac{6}{4} + \frac{1}{4}$ which is the same as

below.

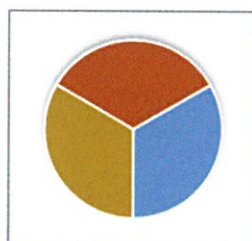
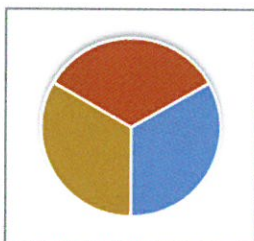
Shade a quarter in the first circle **red**. The amount shaded is shown in the fraction sequence



PART A. Partition the following circles into quarters:

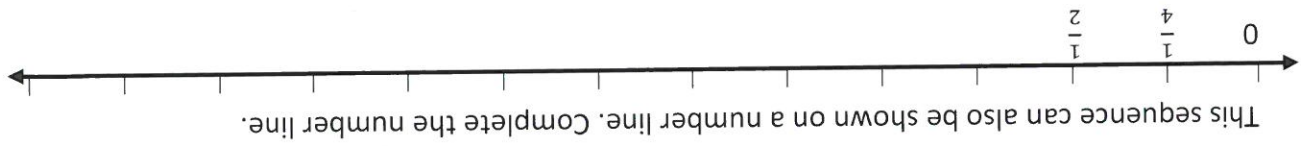
Think: How can we mentally add and subtract fractions?

Whole Class Activity 1



$$2 \div \frac{1}{3} = 6$$

We can solve a problem like $\frac{1}{2} \times \frac{2}{3}$ by visualizing shapes such as circles or rectangles with $\frac{1}{2}$ shaded and then dividing this half into two equal pieces to get $\frac{1}{4}$. We can solve **division** problems by thinking of division as grouping. For example, when solving $2 \div \frac{1}{3}$, we can think of this as 'how many thirds in 2?' We can draw a diagram to represent 2 wholes and partition the wholes into thirds. We can then count how many thirds in 2 wholes.



Why is $\frac{2}{4} = \frac{1}{2}$? Explain using the circles or the number line.

Why is $\frac{4}{6} = 1\frac{1}{3}$? Explain using the circles or the number line.

Nancy, who worked on fractions in Unit 2, thought she could now solve a problem like $1\frac{1}{2} + 1\frac{1}{4}$ by visualising fractional parts in the circles and counting forwards on a number line.

Nancy wondered whether she could solve subtraction problems such $2\frac{1}{4} - \frac{1}{2}$ in a similar way. Explain how Nancy could solve this problem by visualizing the circles.

Explain how Nancy could solve this problem by counting backwards on the number line.

Work with a partner and use your understanding of the number line and the partitioned circles to explain the answers to:

- a) $\frac{1}{4} + \frac{1}{2}$
- b) $1\frac{2}{3} + \frac{4}{3}$
- c) $1\frac{1}{4} + 1\frac{1}{2}$
- d) $\frac{4}{3} - \frac{1}{2}$
- e) $1\frac{1}{4} - \frac{1}{2}$

PART C. How could you solve a problem like $\frac{7}{10} + \frac{2}{1}$? Explain your thinking.

Practice Exercise 1

1. Solve the following addition problems. Use a number line or diagrams to help if necessary.

a) $\frac{1}{2} + 1\frac{1}{5}$

b) $\frac{6}{1} + \frac{3}{1}$

c) $2\frac{1}{1} + \frac{4}{1} + 1$

d) $1\frac{1}{2} + \frac{6}{5}$

e) $\frac{3}{2} + 2\frac{1}{2}$

f) $\frac{10}{3} + 2\frac{2}{1} + \frac{10}{1}$

2. Solve the following subtraction problems. Use a number line or diagrams to help if necessary.

a) $2\frac{1}{1} - \frac{4}{2}$

b) $\frac{6}{5} - \frac{3}{1}$

c) $3\frac{1}{1} - 1\frac{1}{2}$

d) $\frac{3}{2} - \frac{1}{1}$

e) $1\frac{1}{2} - \frac{6}{5} - \frac{6}{1}$

f) $2\frac{10}{7} - 1\frac{1}{1} + \frac{1}{2}$

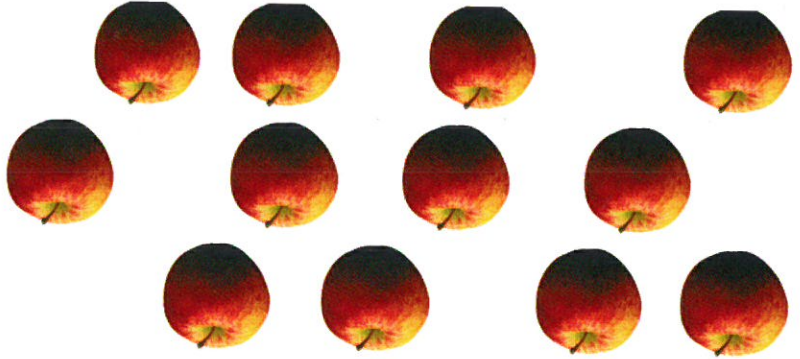
Reflection and Discussion

Nancy's friend Kaleb, solved some easily visualised fractions by thinking of them as decimals. How would Kaleb solve $\frac{1}{4} + 1\frac{1}{2}$ and $2\frac{10}{7} - 1\frac{1}{2}$? Use jottings to show his thinking.

What are the answers to these problems in (i) decimals and (ii) fractions?

What would be the best form to write these answers in? Circle DECIMALS or FRACTIONS. Explain your thinking.

Which of the problems in PRACTICE EXERCISE 1, numbers 1 and 2 could be solved using Kaleb's method?



Nancy had a bag of 12 apples.

Think: How can we mentally multiply with fractions?

Whole Class Activity 2

4. Solve the following problems.
- If John loses $\frac{1}{2}$ kg more, he will weigh the same as Peter who weighs $79\frac{3}{4}$ kg. How much does John weigh?
 - Bob the butcher uses $3\frac{1}{2}$ kg of mince to make hamburgers on Friday and $\frac{6}{5}$ kg less for the hamburgers on Saturday. How many kilograms of hamburger does he make on Saturday?
 - John runs for $1\frac{10}{11}$ hours. If Riccardo had run half an hour more, he would have run the same as John. How long did Riccardo run for?
 - Julie teaches ballet for $2\frac{4}{3}$ hours on a Wednesday, $1\frac{1}{2}$ hours on a Saturday and $\frac{1}{2}$ hour on a Sunday. How many hours does she teach ballet altogether?
 - Chef Nellie uses $1\frac{1}{4}$ cups of milk for her scones and $\frac{4}{3}$ cup less for her biscuits. How much milk does she use for her biscuits?
 - Laurie cuts off one $2\frac{3}{4}$ m piece and one $\frac{1}{2}$ m piece from a 6 m length of rope. How much rope does Laurie have left?

$\frac{4}{3} + 3\frac{1}{2} = 4\frac{1}{1}$	$2\frac{10}{3} + \frac{1}{2} = 2\frac{4}{5}$	$1\frac{10}{9} + \frac{1}{2} = 2\frac{5}{2}$	$2\frac{1}{2} - \frac{1}{6} = 2\frac{1}{3}$	$\frac{1}{5} + 2\frac{1}{10} = 2\frac{3}{10}$
$3\frac{1}{4} - \frac{1}{2} = 3\frac{1}{2}$	$\frac{3}{2} + \frac{1}{2} = 1\frac{6}{2}$	$\frac{1}{10} + 2\frac{1}{2} = 3\frac{1}{10}$	$2\frac{1}{4} - \frac{1}{2} = 2\frac{1}{4}$	$\frac{7}{10} - \frac{1}{2} = \frac{1}{5}$
$1\frac{4}{4} - \frac{1}{2} = \frac{4}{3}$	$\frac{6}{5} + \frac{1}{3} = 1\frac{6}{15}$	$\frac{10}{9} + 2\frac{1}{2} = 2\frac{10}{12}$	$\frac{10}{9} - \frac{1}{10} = \frac{5}{4}$	$\frac{3}{2} + 1\frac{1}{2} = 1\frac{6}{2}$

3. The following table has 4 incorrect number sentences. Cross out the incorrect problems.

She wanted to know what $\frac{1}{4}$ of 12 apples was ($\frac{1}{4} \times 12$). Nancy shared the apples into 4 equal groups. Draw circles around the apples to show how she did this.

How many apples is $\frac{1}{4}$ of the bag?

To find $\frac{4}{3}$ of the apples, Nancy calculated how many apples were in 3 of the circled groups. How many apples are $\frac{4}{3}$ of the bag ($\frac{4}{3} \times 12$)?

Next Nancy wanted to know how many apples is $\frac{3}{1}$ of the bag ($\frac{3}{1} \times 12$). She shared the apples into 3 equal groups. Use a different colour and draw circles around the apples to show how she did this.

How many apples is $\frac{1}{3}$ of the bag?

How many apples are $\frac{2}{3}$ of the bag ($\frac{2}{3} \times 12$)?

Draw a collection of 16 items, eg, flowers, balloons, emojis.

Use the items to find answers to:

1. $\frac{4}{3} \times 16$

2. $\frac{8}{5} \times 16$

3. $1\frac{1}{4} \times 16$

Explain to a partner how you would you calculate $\frac{10}{3} \times 210$? $\frac{5}{4} \times 150$? $1\frac{1}{2} \times 10$? Solve the problems. What jottings (if any) did you use to help you?

Nancy cut an apple in half. She wants to find $\frac{1}{4}$ of $\frac{1}{2}$ ($\frac{1}{4} \times \frac{1}{2}$).

She draws a full circle to represent an apple and shades half. She then draws lines in the half to divide it into quarters. She knows how to partition the whole apple in order to say how much of an apple is $\frac{1}{4}$ of $\frac{1}{2}$.



How much apple is $\frac{1}{4}$ of $\frac{1}{2}$?

Use diagrams to find how much apple is:

1. $\frac{1}{2}$ of $\frac{1}{2}$

2. $\frac{3}{4} \times \frac{1}{2}$

3. $\frac{1}{4} \times \frac{1}{3}$

Explain to a partner how you would you calculate $\frac{1}{2} \times \frac{5}{4}$? $\frac{1}{4} \times \frac{4}{1}$? $6\frac{1}{2} \times \frac{2}{1}$? Solve the problems. What jottings (if any) did you use to help you?

Practice Exercise 2

1. Solve the following problems

a) $\frac{5}{1} \times 25$

b) $\frac{3}{5} \times 25$

c) $\frac{4}{3} \times 84$

d) $\frac{10}{7} \times 220$

e) $16 \times 2\frac{1}{2}$

f) $1\frac{4}{1} \times 240$

2. Solve the following problems

a) $\frac{1}{2} \times \frac{4}{1}$

b) $\frac{3}{1} \times \frac{4}{1}$

c) $\frac{2}{1} \times \frac{4}{3}$

e) $\frac{1}{2} \times 1\frac{1}{2}$

f) $10\frac{2}{1} \times 1\frac{1}{2}$

3. Solve the following everyday problems.

a) Kerry earns \$440 per week. She budgets $\frac{4}{3}$ of this on expenses. What are Kerry's expenses each week?

b) Pia has a 120 m reel of copper. She uses $\frac{3}{2}$ of it on an art project. How much copper does Pia use?

c) Bob works $1\frac{1}{2}$ hours each week for 18 weeks. How many hours does Bob work in total?

d) A mixture for making playdough needs $2\frac{1}{2}$ cups of salt. Patricia wants to make $\frac{1}{2}$ of this recipe. How much salt does she need?

e) Nick pours $\frac{5}{2}$ of a 250 mL cup of milk into a bowl. He then adds another 50 mL. How much milk does he pour?

- f) Shanna is making a salad dressing but she wants to make half of the recipe. The recipe uses $\frac{2}{3}$ cup of oil and $\frac{1}{4}$ cup vinegar. How much oil does she need? How much vinegar?

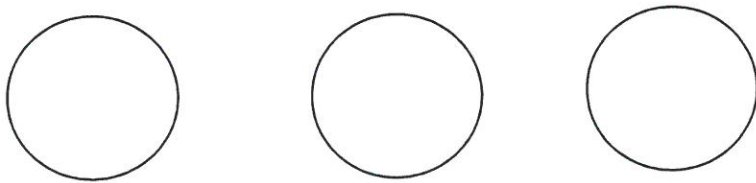
Whole Class Activity 3

Think: How can we mentally divide with fractions?

When solving problems like $3 \div \frac{1}{4}$, we can think of it as 'how many quarters in 3?'

Work with a partner to show how many quarters in 3 using:

A. Circles to represent 3 whole values:



B. A number line:



Now draw (i) 6 circles to represent 6 wholes (ii) a number line from 0 to 6.

Use the diagrams to find answers to:

1. $6 \div \frac{1}{2}$

2. $6 \div \frac{3}{4}$

3. $6 \div 1\frac{1}{2}$

Explain to a partner how you would you calculate: $7 \div \frac{1}{3}$? $6\frac{1}{2} \div \frac{1}{4}$? Solve the problems. What jottings (if any) did you use to help you?



$1\frac{1}{2} \div \frac{1}{2}$	$1\frac{1}{2} \times \frac{1}{2}$
$1\frac{3}{8} - \frac{1}{2}$	$\frac{3}{2} \times 21$
	$4 \div \frac{1}{5}$
	$2\frac{1}{3} - 1\frac{2}{3} + \frac{1}{6}$
	$1\frac{1}{2} \times 6$

1. Use a diagram, number line or other jottings to solve the following problems.

Reflection on Learning

2. Solve the following everyday problems.
- Bruce cuts a 7 m long piece of rope into $\frac{3}{4}$ m lengths. How many pieces did he cut?
 - A rectangle has an area of 15 cm^2 . If its width is $\frac{4}{3} \text{ cm}$, what is its length?
 - If Chef Jake packaged 18 kg of meat into $1\frac{1}{2}$ kg bags, how many bags of meat would he have?
 - In the American money system, 25c is called a quarter. How many quarters are in \$2?
 - Travis buys $1\frac{1}{2}$ kg of nuts for \$9. What is the price of the nuts per kilogram?
 - How many 50 cents are in \$7.50? (HINT: It may help to think of 50c as $\frac{1}{2}$ dollar and \$7.50 as $7\frac{1}{2}$).

1. Solve the following division problems. Use a diagram or number line to help if necessary.
- $4 \div \frac{2}{3}$
 - $5 \div \frac{1}{4}$
 - $6 \div \frac{1}{3}$
 - $3 \div \frac{4}{3}$
 - $2\frac{1}{5} \div \frac{1}{5}$
 - $12 \div 1\frac{1}{2}$

Practice Exercise 3

2. Solve the following everyday problems.

- Sophie has a mobile phone plan. The plan costs \$60 per month. Sophie pays $\frac{4}{3}$ of the bill and her Mum pays the rest. How much does Sophie pay?
- Sophie makes $\frac{3}{2}$ of her phone calls between 9 am and noon. She makes $\frac{6}{1}$ of the calls between noon and 3 pm. What fraction of her calls is made between 9 am and 3 pm? What fraction is made between 3 pm and 9 am?
- Sophie talks to Ella on her mobile for $5\frac{4}{3}$ mins. She talks to Paul for $4\frac{1}{2}$ mins. How much longer did she talk to Ella than Paul? Sophie talks to Ella for twice as long the next day. How long was this phone call?
- Sophie is voting for 'The Y Factor' winner on her mobile phone. Each vote takes $\frac{1}{2}$ min to record. How many votes can Sophie make continuously in 10 minutes?

OLNA Practice Questions

- Last year, the number of people that went to the 'Groove in the South' music festival was 14 000. The number of people who went to the music festival this year was $\frac{5}{7}$ of last year's figure. How many people went to the festival this year?
- What is the number exactly halfway between $2\frac{4}{3}$ and $3\frac{3}{4}$?

- A. 2000 B. 10 000 C. 9 500 D. 12 000

Topic 3

Using Mental Strategies to Solve Problems Involving Decimals

Mathematics Discussion

In Units 1 and 2 of the Foundations Mathematics course, we used mental strategies such as place value, partitioning and our knowledge of basic facts to solve problems involving whole numbers, money and the four operations. We used written jottings to perform calculations we could not completely store in our heads. These jottings included diagrams, number lines or number calculations. For problems involving 'difficult' numbers, we used a calculator.

In this topic, we will work through exactly the same decision making, except we are now extending our work with whole numbers to include problems involving decimals.